

ZERODHA CLONE - PROJECT INFORMATION

Full-Stack Stock Trading Platform

1. Project Overview

This project is a comprehensive Full-Stack clone of Zerodha, India's leading stock broker platform. It provides a complete trading ecosystem divided into three distinct modules:

- Marketing Website (Frontend): Zerodha landing, pricing, products, and support pages.
- Trading Console (Dashboard): Kite clone with real-time watchlist, buy window, holdings, and funds.
- Backend Engine (REST API): Node.js & Express API with MongoDB for order and portfolio persistence.

2. Technology Stack & Architecture

Module	Technology Stack	Key Features
Frontend		
Dashboard		
Backend		
Database		

3. Core Features & Capabilities

- [+] Market Watchlist: Dynamic stock listings with NIFTY/SENSEX indices and % change.
- [+] Buy Action Modal: Order placement window allowing custom stock quantities and prices.
- [+] Holdings Management: Portfolio view with investment value, current value, and P&L.
- [+] Open Positions: Real-time intraday MIS and CNC position tracking.
- [+] Funds Breakdown: Instant margin calculations for equity and commodity segments.
- [+] Interactive Charts: Asset allocation doughnut charts and vertical performance bars.

4. Backend REST API Endpoints

Endpoint	Method	Function & Description
/allHoldings		
/allPositions		
/allOrders		
/newOrder		

5. Database Schemas (Mongoose)

- Holdings Schema: name (String), qty (Number), avg (Number), price (Number), net (String), day (String)
 - Positions Schema: product (String), name (String), qty (Number), avg (Number), price (Number), net, day
 - Orders Schema: name (String), qty (Number), price (Number), mode (String: 'BUY'/'SELL')
- This application provides an authentic, modular recreation of Zerodha's trading infrastructure. It demonstrates end-to-end full stack proficiency including state management, API design, database modeling, and interactive data visualizations.

6. How to Run the Project Locally

Step 1: Start Backend

```
cd backend --> npm install --> npm start (Runs at http://localhost:3002)
```

Step 2: Start Dashboard

```
cd dashboard --> npm install --> npm start (Runs at http://localhost:3001)
```

Step 3: Start Marketing Frontend

```
cd frontend --> npm install --> npm start (Runs at http://localhost:3000)
```

Summary Note: